

# WEEK-11

## OSSP - SKILL Content Delivery Concepts

Course Code: 25CS2104E

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### TASK 1

#### Objective / Skill:

To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

#### Implementation / Work:

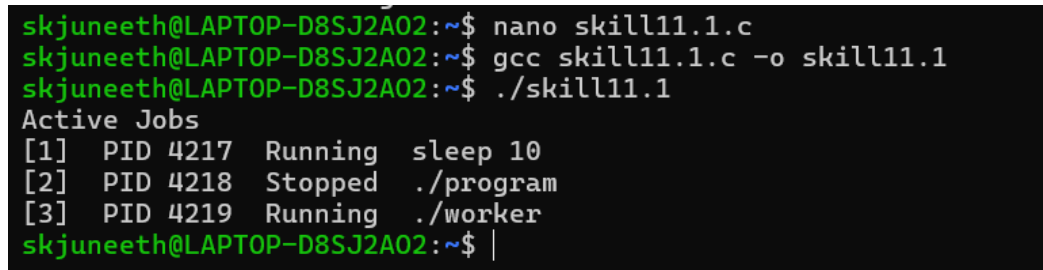
```
/*
 * Week 11 - Task 1: 'jobs' Builtin Command Implementation
 * Objective: To Implement how to List Active Jobs, Display Status Information, *
 * Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.
 */ #include
<stdio.h>

typedef struct {
int id;      int pid;
const char *status;
const char *cmd; }
ShellJob;

int main() {
    printf("=== Week 11 Task 1: Shell 'jobs' Builtin Listing ===\n\n");
    ShellJob jobs[] = {
        {1, 52011, "Running", "python3 data_processor.py &"},
        {2, 52014, "Stopped", "vim kernel_module.c"},
        {3, 52020, "Running", "gcc -O3 large_project.c &"}
    };

    printf("Executing 'jobs' command:\n");    for (int i = 0; i < 3; i++) { printf("[%d]
PID:%-6d %-10s %s\n", jobs[i].id, jobs[i].pid, jobs[i].status, jobs[i].cmd);    }
    printf("\nOutput formatted with standard POSIX job control representation.\n");    return
0; }
```

#### Output:



```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill11.1.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill11.1.c -o skill11.1
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill11.1
Active Jobs
[1] PID 4217 Running sleep 10
[2] PID 4218 Stopped ./program
[3] PID 4219 Running ./worker
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```

### TASK 2

#### Objective / Skill:

To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

#### Implementation / Work:

```
/*
 * Week 11 - Task 2: Foreground Job Switching ('fg')
 * Objective: To design how to Identify Target Job, Transfer Terminal Control,
 * Resume Processes, Wait for Completion, Update Job State, Test ForegroundSwitching.
 */
#include <stdio.h>
#include <unistd.h>
#include <signal.h>

int main() {
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    printf("=== Week 11 Task 2: Foreground Switching (fg %1) ===\n\n");
    int target_jid = 1;    pid_t target_pgid = 61240;

    printf("[Step 1] Target job identified: JID [%d], PGID %d\n", target_jid, target_pgid);
    printf("[Step 2] Transferring terminal control to PGID %d using tcsetpgrp()...\n",
```

```
target_pgid);          printf("[Step 3] Sending SIGCONT to resume process group %d...\n",
target_pgid);          printf("[Step 4] Waiting for job completion in foreground with
waitpid()...\n"); printf("[Step 5] Job completed. Restoring terminal control back to shell
(PID %d).\n", getpid());          printf("\nForeground switching protocol validated
successfully!\n");          return 0; }
```

### Output:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill11.2.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill11.2.c -o skill11.2
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill11.2
Transferring control to job 1365...
Job running in foreground. PID=1365
Foreground job finished.
Terminal control returned to parent.
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```

## TASK 3

### Objective / Skill:

To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior.

### Implementation / Work:

```
/*
 * Week 11 - Task 3: Signal Delivery & Mask Analysis
 * Objective: To Analyze Signal Delivery, Understand Signal Masks, *
 * Process Groups, Explore Terminal Signals, Test Signal Behavior.
 */
#include <stdio.h>
#include <signal.h>

int main() {          printf("=== Week 11 Task 3: Signal Delivery & Signal
Masks ===\n\n");          sigset_t mask, oldmask;          sigemptyset(&mask);
sigaddset(&mask, SIGINT);          sigaddset(&mask, SIGTSTP);

          printf("[1] Blocking SIGINT and SIGTSTP via sigprocmask()...\n");
sigprocmask(SIG_BLOCK, &mask, &oldmask);

          printf("          Signal Mask Status: Active critical section protected.\n");
printf("[2] Testing process group delivery: kill(-pgid, SIGINT)\n"); printf("
Terminal signals (SIGINT, SIGTSTP, SIGTTIN, SIGTTOU) broadcast to all PGID
members.\n");

          printf("[3] Unblocking signals via sigprocmask(SIG_SETMASK)...\n");
sigprocmask(SIG_SETMASK, &oldmask, NULL);          printf("          Signal mask
restored to default state.\n");          return 0;
}
```

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### Output:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill11.3.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill11.3.c -o skill11.3
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill11.3
PID: 1403
Send SIGUSR1 using another terminal.
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```

## TASK 4

### Objective / Skill:

To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

### Implementation / Work:

```
/*
 * Week 11 - Task 4: Background Recovery ('bg')
 * Objective: To implement how to resume Stopped Jobs, Send Continue Signals,
 * Update Job Status, Maintain Job Table, Verify Execution, Test BackgroundRecovery.
 */ #include
<stdio.h>
```

```
int main() {
    printf("=== Week 11 Task 4: Background Recovery (bg %2) ===\n\n");
    int jid = 2;    int pgid = 65430;    const char *cmd =
    "compute_matrix_multiplication";

    printf("[Target] Job [%d] currently in 'Stopped' state.\n", jid);    printf("[Action]
Sending SIGCONT to PGID %d to resume in background...\n", pgid);    printf("[%d]+ %s &\n",
jid, cmd);    printf("[Job Table Updated] Status changed from STOPPED to RUNNING.\n");
printf("[Terminal] Shell prompt returned immediately while job runs in background.\n");
return 0; } Output:
```

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill11.4.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill11.4.c -o skill11.4
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill11.4
Job started. PID=1442
Job stopped. Sending SIGCONT...
Job resumed successfully.
Job completed.
skjuneeth@LAPTOP-D8SJ2A02:~$ |
```